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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 06/22/2005 |
| **PI Name:** | Bloomfield, Susan A. Ph.D. | | |
| **Project Title:** | Bone and Muscle Recovery from Simulated Microgravity | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Bone Loss Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 77843-4375 | **Congressional District:** | 17 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | NSBRI |
| **Start Date:** | 01/01/2001 | **End Date:** | 02/29/2004 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 3 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 2 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Smith, Carolyn ( Baylor College of Medicine )<br>Hogan, Harry ( Texas A&M University ) | | |
| **Grant/Contract No.:** | NCC 9-58-BL00205 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | In spite of continued development of countermeasures for bone loss, ISS astronauts still exhibit loss of bone mineral at the spine and hip upon their return. Published data suggest that, among those humans losing bone during spaceflight, recovery of bone mineral is extremely slow; there are no published data confirming that all bone mineral lost by humans during long duration spaceflight is recovered over time, though anecdotal reports suggest that after 2-3 years bone mass has recovered. By contrast, muscle mass and strength generally recover quite quickly following return to normal weightbearing activity after spaceflight. Given that bone mineral density (BMD) is highly predictive of bone strength, there will be some period during recovery after muscle strength has normalized during which muscle strength will far exceed usual (pre-flight) bone strength. Our objectives in this project were to use the well-validated model of the hindlimb unloaded (HU) rat to perform more invasive studies than is feasible in humans to determine the time course of recovery of functional properties in a muscle-bone pair after a period of simulated microgravity, and to test the | | |

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| **Task Description:** | effectiveness of mechanical and pharmacological interventions in promoting return of bone strength without exacerbating a muscle-bone strength mismatch. The time course studies comprehensively describe the recovery of muscle and bone following 28 days of HU, with in vivo measures of bone density and geometry and of muscle functional properties. Ex vivo analyses of bone marrow osteoprogenitor cells and histomorphometric analyses and mechanical testing of cortical and cancellous bone sites were completed. Key findings include: functional properties of the plantarflexor muscles recovered completely within 14 days of return to normal weightbearing; normal weightbearing alone stimulated no recovery of volumetric bone mineral density (vBMD) at the proximal tibia until between 56 and 84 days of recovery, resulting in a significant mismatch between the two tissues strength between 14 and 56 days of recovery; and cell culture studies of marrow from femurs of rats recovering from HU suggest a decline in mature osteoblast activity in spite of greater numbers of bone marrow osteoprogenitor cells. Future research should better define the clinical relevance of this mismatch between functional properties of muscle-bone pairs during recovery from prolonged disuse, with a particular emphasis on the risk of avulsion fracture.<br>Two interventions were tested during a 28-d recovery period from HU for their ability to stimulate a more rapid recovery of bone with the goal of minimizing the mismatch between bone and muscle strength. Two groups of rats were implanted with tibial nerve cuffs so that stimulated isometric contractions of the plantar flexors could be performed during recovery, mimicking high-intensity resistance training. The high-force muscle stimulation protocol during recovery was only partially effective in enhancing recovery of total and cancellous VBMD at the proximal tibia vs those rats with nerve cuffs that received no training. There was a trend ($p = .07$) for peak isometric torque of the posterior crural muscles to be greater after 28 days of muscle stimulation as compared to untreated rats. Downregulation of muscle mass and strength in all nerve-cuff implanted rats complicates interpretation of some of these data. On the other hand, our intervention with an FDA-approved osteoporosis treatment, intermittent PTH, effectively stimulated rapid increases in bone mass at the proximal tibia during recovery, at a rate nearly equal to that for muscle strength. Micro-CT studies documented higher cancellous bone volume in the proximal tibia of PTH-treated rats versus controls, accounted for by higher trabecular thickness and number, and reduced trabecular spacing. Material properties of the proximal tibia cancellous compartment exhibited parallel improvements. The mechanism for these improvements stems from the stimulation of osteoblast activity by PTH. Bone formation rates were increased on cancellous and cortical bone surfaces at proximal and mid-shaft tibia. Hence both cortical and cancellous bone compartments can respond to the anabolic effects of intermittent PTH treatment after a prolonged period of mechanical unloading. Because several parameters of bone mass and volume were increased well beyond baseline values after 28 days of PTH treatment (using 80 μg/kg BW/d), it is likely that a much smaller dose of PTH would be effective in hastening bone recovery from HU and post-spaceflight. Further work should clarify how low a dose would be effective in stimulating this more rapid return of bone mass, and also whether these gains in bone mass and strength can be retained after short-term PTH treatment is discontinued. Such maintenance may well require the alternative anabolic stimulus of regular exercise. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | These results describing a serious mismatch between muscle and bone strength should be of high interest to rehabilitation researchers and practitioners working with patients exposed to prolonged bed rest or immobilization of a limb/joint after injury. Given that muscle responds fairly quickly to simple remobilization, according to these results, practitioners need to heed the risk for bony injury during rehabilitation, particularly if it includes aggressive resistance training that will impose high magnitude forces on a weakened bone site. Previous work by Scandinavian researchers (Sievanen et al., Bone 1996) estimated a 3-fold increase in strain on bone during quadriceps contractions after 4 months immobilization of the knee joint following a complicated ACL injury. This was a timepoint some six months into rehabilitation, when quadriceps muscle strength had fully recovered, but only 50% of lost BMD had been recouped. Existing literature suggests that, in young healthy individuals, resistance training and even impact loading (such as plyometrics) is most effective in stimulating increases in BMD. Such recommendations should be adapted when rehabilitating individuals who have experienced significant bone loss, to minimize very high-magnitude strains on the affected bone. Rehabilitation strategies found to be effective with returning astronauts could and should be adapted for patients having endured Earth-bound immobilization.<br>Despite evidence in previous studies for depressed osteoblast function after disuse, it appears that, once weightbearing is resumed, cancellous bone is highly responsive to the anabolic stimulus provided by intermittent PTH. Short-term treatment with intermittent PTH may prove beneficial after prolonged bed rest or unavoidable limb immobilization in accelerating recovery of cancellous VBMD and architecture and restoring normal BFR in cortical bone. Careful dose-response studies should be performed to optimize the dose of intermittent PTH, as the dose used with this small animal model more than replaced the bone lost. Hence, much lower doses of intermittent PTH may be effective. Further research will be needed to define whether PTH-induced gain in bone can be effectively maintained once PTH treatments are withdrawn; it is likely that a continued anabolic stimulus, such as regular weightbearing exercise, will be required. |
| **Task Progress:** | We have made solid progress this final year of the project on finishing the most of the animal experiments described in our original proposal. It should be noted that significant budget reductions incurred in the second year of this project forced a reduction in the number of interventions we could study during recovery. In addition, considerable difficulty was encountered in negotiating a material transfer agreement with Genentech to acquire human recombinant growth hormone. Hence we chose to drop the growth hormone arm of the pharmacological interventions and concentrate on that intervention (intermittent PTH) that had the best promise of impacting directly on bone recovery per se. We were best equipped to pursue the high-force muscle contraction protocol as a mechanical intervention and proceeded with this option as the most strategic choice, since resistance training (which this protocol attempts to mimic) is currently regarded as an important countermeasure for human astronauts. The restricted funding also delayed our ability to complete histological analyses from tissues collected from these animal experiments, so our efforts at getting all results to publication stage has been slowed. In addition to the publications listed in this report, we have data pending for at least 3 more manuscripts. One of those should be submitted by January 20 of 2005, and the other two by the end of February 2005.<br>We established a successful collaboration with Dr Dianna Cody's lab at MD Anderson Cancer Center in Houston, TX, to complete microCT studies of a subset of bones in the intermittent PTH studies. These data were likely pivotal in our securing an oral presentation at the American Society for Bone and Mineral Research meetings in 2003 (only top 10-15% of submitted abstracts are selected for oral presentation). Year 3 also saw the completion of studies using implanted nerve cuffs to allow for our "resistance training" intervention during recovery. We documented a full recovery of muscle strength (to pre-unloading values) in those rats subjected to this training regimen; however, both muscle mass |

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| | and peak torques in the trained rats were lower than those in rats without implantednerve cuffs. Some of our interpretation of these data may be made difficult by this down regulation of muscle mass/strength in these rats. Also, a longer training period may be required for a positive effect, which has been demonstrated in a study with older Fischer rats (Lowe et al, J. Appl. Physiol. 2004) Histological and microcrack evaluations of the tibiae from rats subjected to the muscle stimulation protocol are on-going. Our proven ability to use this technique did allow us to successfully compete for funding renewal, using this muscle stimulation protocol as one of two exercise interventions to be tested during hindlimb unloading in combination with an anti-resorptive pharmacological agent (alendronate). |
| **Bibliography Type:** | Description: (Last Updated: 01/20/2026) |
| **Articles in Peer-reviewed Journals** | Narayanan R, Allen MR, Gaddy D, Bloomfield SA, Smith CL, Weigel NL "Differential skeletal responses of hindlimb unloaded rats on a Vitamin D deficient diet to 1,25-dihydroxyvitamin D and its analog, Seocalcitol (EB1089)." Bone. 2004 Jul;35(1):134-43. PMID: 15207749 , Jul-2004 |
| **Articles in Peer-reviewed Journals** | Warren GL, Stallone JL, Allen MR, Bloomfield SA. "Functional recovery of the plantarflexor muscle group after hindlimb unloading in the rat." Eur J Appl Physiol. 2004 Oct;93(1-2):130-8. PMID: 15248071 , Oct-2004 |
| **Articles in Peer-reviewed Journals** | Allen MR, Hogan HA, Bloomfield SA. "Differential bone and muscle recovery following hindlimb unloading in skeletally mature male rats." J Musculoskelet Neuronal Interact. 2006 Jul-Sep;6(3):217-25. PMID: 17142941 , Jul-2006 |
| **Awards** | Hogan HA. "To H.A. Hogan: Ruth & William Neely '52/Dow Chemical Faculty Fellow, Look College of Engineering, January 2004." Jan-2004 |
| **Awards** | Bloomfield SA. " To S.A. Bloomfield, February 2004: Honor Award, from Texas Regional Chapter of the American College of Sports Medicine. " Feb-2004 |
| **Dissertations and Theses** | Allen MR. "Mechanisms of impaired osteoblast function during disuse. " Dissertation, August 2003. , Aug-2003 |
| **Dissertations and Theses** | Lemmon HB. "Methods for reduced platen compression (RPC) test specimen cutting location using micro-CT and planar radiographs." Dissertation, August 2003. , Aug-2003 |
| **Dissertations and Theses** | Ramaswamy R. "Correlating Mechanical Properties of Cancellous Bone in the Rat with Various Density Measures." Dissertation, December 2003. , Dec-2003 |
| **Presentation** | Allen, M. R., S. A. Bloomfield. "Mechanisms for reduced osteoblast activity with unloading of bone. " May-2003 |
| **Presentation** | Allen, M. R..S. A. Bloomfield. "Absence of mechanical load results in an attenuation of marrow ablation-induced increase in cancellous bone formation rate. " Sep-2003 |
| **Presentation** | Baek, K., A. Currado, M. Allen, S. Bloomfield. "Effect of food restriction and hindlimb unloading on serum leptin and histomorphometric measures of cancellous bone. " Oct-2004 |
| **Presentation** | Baek, K., A. Currado, M. R. Allen, S. A. Bloomfield. "Comparison of serum leptin concentration under simulated microgravity condition or food restriction or combination of two. " Sep-2003 |
| **Presentation** | Baek, K., J. L. Stallone, S. A. Bloomfield. "Beta-blockade mitigates bone loss with hindlimb unloading. " Apr-2004 |
| **Presentation** | Bloomfield, S. A. "Differential recovery of bone and muscle after prolonged disuse: do we need a new strategy? " Jan-2004 |
| **Presentation** | Bloomfield, S. A. "How animal research has informed the new ACSM position stand on bone health and osteoporosis " May-2004 |
| **Presentation** | Bloomfield, S. A., K. Baek, J. L. Stallone, M. R. Allen. "Intermittent PTH during recovery replaces cancellous bone loss during hindlimb unloading by increasing bone formation rate. " Apr-2004 |
| **Presentation** | Bloomfield, S. A., M. R. Allen, H. Lemon, and H. A. Hogan "PTH replaces cancellous bone lost during 28 days of hindlimb unloading in skeletally mature rats. " Sep-2003 |
| **Presentation** | Bloomfield, S. A., M. R. Allen. "Pronounced mismatch of bone density and Muscle strength during recovery from long-term hindlimb unloading. " May-2003 |
| **Presentation** | Hogan, H. A., D. Older, M. Rose, S. Vaculik, and M. R. Allen "Loading direction affects mechanical property results in bending of rat tibia specimens " Oct-2004 |
| **Presentation** | Swift, J. M., T. F. Kennedy, S. A. Bloomfield "Intermittent PTH during recovery from hindlimb unloading: impact on mechanical properties and bone formation rate in cortical bone. " Oct-2004 |